

Atomic Structure

1. Which is the main constituent of the nucleus?

- a) Proton
- b) Neutron
- c) Electron
- d) Positron

Correct answer: b) Neutron

2. The number of protons in the nucleus is called _____.

- a) Atomic number
- b) Mass number
- c) Neutron number
- d) Electron number

Correct answer: a) Atomic number

3. Isotopes have same _____ but different _____.

- a) Atomic number, mass number
- b) Mass number, atomic number
- c) Neutrons, protons
- d) Electrons, protons

Correct answer: a) Atomic number, mass number

4. Atomic mass is approximately equal to _____.

- a) Protons
- b) Neutrons
- c) Protons + neutrons
- d) Electrons

Correct answer: c) Protons + neutrons

5. The maximum number of electrons in n shell is _____.

- a) $2n$
- b) $2n^2$
- c) n^2
- d) $8n$

Correct answer: b) $2n^2$

6. K shell can have maximum _____ electrons.

- a) 2
- b) 8
- c) 18
- d) 32

Correct answer: a) 2

7. Valency of elements is determined by _____.

- a) Protons
- b) Neutrons
- c) Valence electrons
- d) Atomic mass

Correct answer: c) Valence electrons

8. Electronic configuration of sodium (11) is _____.

- a) 2,8,1
- b) 2,8,2
- c) 2,8,3
- d) 2,1,8

Correct answer: a) 2,8,1

9. The octet rule refers to _____.

- a) 8 protons
- b) 8 neutrons
- c) 8 valence electrons
- d) 8 shells

Correct answer: c) 8 valence electrons

10. Molecule of water has _____ atoms.

- a) 1 oxygen, 2 hydrogen
- b) 2 oxygen, 1 hydrogen
- c) 3 oxygen
- d) 2 hydrogen

Correct answer: a) 1 oxygen, 2 hydrogen

11. Atomicity of helium is _____.

- a) 1

b) 2

c) 3

d) 4

Correct answer: a) 1

12. Which has highest atomicity?

a) Glucose

b) Water

c) Helium

d) Oxygen

Correct answer: a) Glucose

13. Molecular mass of oxygen (O_2) is _____.

a) 16 u

b) 32 u

c) 18 u

d) 1 u

Correct answer: b) 32 u

14. Formula mass of $Ca_3(PO_4)_2$ is _____.

a) 87 u

b) 120 u

c) 310 u

d) 164 u

Correct answer: c) 310 u

15. Avogadro number is _____.

a) 6.023×10^{23}

b) 6.023×10^3

c) 6.023×10^{24}

d) 6.023

Correct answer: a) 6.023×10^{23}

16. 1 mole of any substance contains _____ particles.

a) 6.023×10^{23}

b) 6.023×10^{22}

c) 1×10^{23}

d) 12×10^{23}

Correct answer: a) 6.023×10^{23}

17. Molar mass of water is _____.

a) 18 g

b) 16 g

c) 32 g

d) 1 g

Correct answer: a) 18 g

18. Number of moles in 32 g of O_2 is _____.

a) 1

b) 2

c) 16

d) 32

Correct answer: a) 1

19. 44 g of CO_2 contains _____ moles.

a) 1

b) 2

c) 12

d) 44

Correct answer: a) 1

20. Mass of 0.5 mole of N_2 is _____.

a) 7 g

b) 14 g

c) 28 g

d) 1 g

Correct answer: b) 14 g

21. Proton was discovered by _____.

a) J.J. Thomson

b) Rutherford

c) Goldstein

d) Chadwick

Correct answer: c) Goldstein

22. Neutron was discovered by _____.

- a) Chadwick
- b) Bohr
- c) Dalton
- d) Thomson

Correct answer: a) Chadwick

23. Electron was discovered by _____.

- a) J.J. Thomson
- b) Rutherford
- c) Bohr
- d) Dalton

Correct answer: a) J.J. Thomson

24. Atomic model with electrons in orbits is by _____.

- a) Dalton
- b) Thomson
- c) Rutherford
- d) Bohr

Correct answer: d) Bohr

25. Plum pudding model is by _____.

- a) Thomson
- b) Rutherford
- c) Bohr
- d) Dalton

Correct answer: a) Thomson

26. Isobars have same _____.

- a) Atomic number
- b) Mass number
- c) Electrons
- d) Protons

Correct answer: b) Mass number

27. Isotopes of hydrogen are _____.

- a) Protium, deuterium, tritium

b) H-1, H-2

c) D, T

d) All

Correct answer: a) Protium, deuterium, tritium

28. Carbon-12 and Carbon-14 are _____.

a) Isotopes

b) Isobars

c) Isotones

d) Isomers

Correct answer: a) Isotopes

29. L shell maximum electrons _____.

a) 2

b) 8

c) 18

d) 32

Correct answer: b) 8

30. M shell maximum electrons _____.

a) 8

b) 18

c) 32

d) 2

Correct answer: b) 18

31. Electronic configuration of chlorine (17) _____.

a) 2,8,7

b) 2,8,8

c) 2,7,8

d) 2,8,1

Correct answer: a) 2,8,7

32. Valency of oxygen is _____.

a) 1

b) 2

c) 3

d) 4

Correct answer: b) 2

33. Valency of aluminium is _____.

a) 1

b) 2

c) 3

d) 4

Correct answer: c) 3

34. Molecular formula of glucose _____.

a) $C_6H_{12}O_6$

b) $C_{12}H_{22}O_{11}$

c) H_2O

d) CO_2

Correct answer: a) $C_6H_{12}O_6$

35. Atomicity of P_4 is _____.

a) 1

b) 2

c) 4

d) 3

Correct answer: c) 4

36. Which is monoatomic?

a) Helium

b) Oxygen

c) Water

d) Ozone

Correct answer: a) Helium

37. Molecular mass of H_2SO_4 is _____.

a) 98 u

b) 18 u

c) 32 u

d) 44 u

Correct answer: a) 98 u

38. 18 g water has _____ moles.

- a) 1
- b) 2
- c) 18
- d) 0.5

Correct answer: a) 1

39. Number of atoms in 16 g oxygen _____.

- a) 6.023×10^{23}
- b) $2 \times 6.023 \times 10^{23}$
- c) 3.011×10^{23}
- d) 12×10^{23}

Correct answer: a) 6.023×10^{23}

40. 1 u is approximately mass of _____.

- a) Hydrogen atom
- b) 1/12 of C-12
- c) Proton
- d) Electron

Correct answer: b) 1/12 of C-12

41. Nucleus was discovered by _____.

- a) Rutherford
- b) Bohr
- c) Thomson
- d) Dalton

Correct answer: a) Rutherford

42. Mass of atom is mainly due to _____.

- a) Nucleus
- b) Electrons
- c) Protons only
- d) Neutrons only

Correct answer: a) Nucleus

43. Charge of nucleus is _____.

- a) Positive

b) Negative

c) Neutral

d) Zero

Correct answer: a) Positive

44. Isotopes have different _____.

a) Chemical properties

b) Physical properties

c) Valency

d) Electronic configuration

Correct answer: b) Physical properties

45. Which has 2 valence electrons?

a) Sodium

b) Magnesium

c) Aluminium

d) Carbon

Correct answer: b) Magnesium

46. N shell maximum electrons _____.

a) 32

b) 18

c) 8

d) 2

Correct answer: a) 32

47. Inert elements have _____ valence electrons.

a) 1

b) 4

c) 8

d) 2

Correct answer: c) 8

48. Formula of calcium chloride _____.

a) CaCl

b) CaCl₂

c) Ca₂Cl

d) CaCl_3

Correct answer: b) CaCl_2

49. Atomicity of ozone _____.

a) 1

b) 2

c) 3

d) 4

Correct answer: c) 3

50. Molar mass of NaCl _____.

a) 58.5 g

b) 18 g

c) 32 g

d) 44 g

Correct answer: a) 58.5 g

51. 12 g carbon has _____ moles.

a) 1

b) 12

c) 6

d) 2

Correct answer: a) 1

52. Atoms in 1 mole of ammonia (NH_3) _____.

a) $4 \times 6.023 \times 10^{23}$

b) $3 \times 6.023 \times 10^{23}$

c) $17 \times 6.023 \times 10^{23}$

d) 6.023×10^{23}

Correct answer: a) $4 \times 6.023 \times 10^{23}$

53. Dalton's theory was modified by _____.

a) Discovery of electrons

b) Protons

c) Isotopes

d) All

Correct answer: d) All

54. Thomson's model had _____ charge distribution.

- a) Positive only
- b) Negative only
- c) Uniform positive with electrons
- d) Neutral

Correct answer: c) Uniform positive with electrons

55. Rutherford's experiment used _____ particles.

- a) Electrons
- b) Neutrons
- c) Alpha
- d) Protons

Correct answer: c) Alpha

56. Valency of nitrogen _____.

- a) 1
- b) 2
- c) 3
- d) 4

Correct answer: c) 3

57. Electronic configuration 2,8,2 belongs to _____.

- a) Na
- b) Mg
- c) Al
- d) Si

Correct answer: b) Mg

58. Molecular mass of CO₂ _____.

- a) 44 u
- b) 28 u
- c) 32 u
- d) 18 u

Correct answer: a) 44 u

59. Which is not an isobar?

- a) ⁴⁰Ar and ⁴⁰Ca

b) ^{12}C and ^{14}C

c) ^{32}P and ^{32}S

d) ^{235}U and ^{235}Np

Correct answer: b) ^{12}C and ^{14}C

60. 0 shell maximum electrons _____.

a) 0

b) 2

c) 8

d) 18

Correct answer: b) 2

61. Number of valence electrons in phosphorus _____.

a) 3

b) 4

c) 5

d) 2

Correct answer: c) 5

62. Formula mass of Na_2CO_3 _____.

a) 106 u

b) 58 u

c) 44 u

d) 98 u

Correct answer: a) 106 u

63. Mass of 1 atom of C-12 is _____.

a) 12 u

b) 1 u

c) 6 u

d) 2 u

Correct answer: a) 12 u

64. Bohr's model explains _____.

a) Hydrogen spectrum

b) All atoms

c) Nucleus

d) Isotopes

Correct answer: a) Hydrogen spectrum

65. Which has same number of neutrons?

a) Isotopes

b) Isobars

c) Isotones

d) Atoms

Correct answer: c) Isotones

66. 36 g water has _____ molecules.

a) $1 \times 6.023 \times 10^{23}$

b) $2 \times 6.023 \times 10^{23}$

c) $18 \times 6.023 \times 10^{23}$

d) $36 \times 6.023 \times 10^{23}$

Correct answer: b) $2 \times 6.023 \times 10^{23}$

67. Atomic number of element with 2,8,8,1 config _____.

a) 17

b) 19

c) 20

d) 18

Correct answer: b) 19

68. Which is triatomic?

a) Helium

b) CO_2

c) O_2

d) H_2

Correct answer: b) CO_2

69. Percentage of hydrogen in water _____.

a) 11.1%

b) 88.9%

c) 50%

d) 100%

Correct answer: a) 11.1%

70. Mole concept was given by _____.

- a) Avogadro
- b) Dalton
- c) Proust
- d) Gay Lussac

Correct answer: a) Avogadro

71. Mass number minus atomic number gives _____.

- a) Electrons
- b) Protons
- c) Neutrons
- d) Valency

Correct answer: c) Neutrons

72. Electronic configuration of potassium (19) _____.

- a) 2,8,8,1
- b) 2,8,9
- c) 2,8,8,2
- d) 2,8,7

Correct answer: a) 2,8,8,1

73. Valency determined by electrons gained or _____.

- a) Lost
- b) Shared
- c) Both
- d) None

Correct answer: c) Both

74. Atomicity of sulphur (S_8) _____.

- a) 1
- b) 2
- c) 8
- d) 4

Correct answer: c) 8

75. 28 g nitrogen has _____ moles.

- a) 1

- b) 2
- c) 14
- d) 28

Correct answer: a) 1

76. In Rutherford experiment, most particles _____.

- a) Deflected
- b) Passed undeflected
- c) Absorbed
- d) Scattered

Correct answer: b) Passed undeflected

77. Nucleus diameter compared to atom _____.

- a) Same
- b) Larger
- c) Very small
- d) Infinite

Correct answer: c) Very small

78. Thomson's model failed because _____.

- a) No nucleus
- b) No stability
- c) Both
- d) None

Correct answer: c) Both

79. Isotopes used in _____.

- a) Nuclear reactors
- b) Carbon dating
- c) Medical treatment
- d) All

Correct answer: d) All

80. 2 moles of CO_2 weigh _____.

- a) 22 g
- b) 44 g
- c) 88 g

d) 12 g

Correct answer: c) 88 g

81. Number of protons in magnesium (12) _____.

a) 12

b) 24

c) 6

d) 2

Correct answer: a) 12

82. KLMN shells fill order _____.

a) K then L

b) L then K

c) Random

d) M first

Correct answer: a) K then L

83. Dueterium has _____ neutrons.

a) 0

b) 1

c) 2

d) 3

Correct answer: b) 1

84. Tritium atomic number _____.

a) 1

b) 2

c) 3

d) 0

Correct answer: a) 1

85. Formula mass used for _____.

a) Ionic compounds

b) Covalent only

c) Elements

d) Molecules

Correct answer: a) Ionic compounds

86. 1 g atom means _____ mole.

- a) 1
- b) 6.023×10^{23} atoms
- c) Both
- d) None

Correct answer: c) Both

87. Valence shell incomplete except noble gases have valency _____.

- a) 0
- b) 1-7
- c) 8
- d) Variable

Correct answer: b) 1-7

88. Molecular mass of CH_4 _____.

- a) 12 u
- b) 16 u
- c) 28 u
- d) 18 u

Correct answer: b) 16 u

89. Argon electronic configuration _____.

- a) 2,8,8
- b) 2,8,1
- c) 2,8,2
- d) 2,8,7

Correct answer: a) 2,8,8

90. Which defines element?

- a) Atomic number
- b) Mass number
- c) Neutrons
- d) Electrons

Correct answer: a) Atomic number

91. 0.1 mole oxygen molecules weigh _____.

- a) 1.6 g

b) 3.2 g

c) 16 g

d) 32 g

Correct answer: b) 3.2 g

92. Atoms in one mole of helium _____.

a) 6.023×10^{23}

b) $2 \times 6.023 \times 10^{23}$

c) $4 \times 6.023 \times 10^{23}$

d) 1×10^{23}

Correct answer: a) 6.023×10^{23}

93. Proton charge _____.

a) +1

b) -1

c) 0

d) +2

Correct answer: a) +1

94. Electron charge _____.

a) +1

b) -1

c) 0

d) Variable

Correct answer: b) -1

95. Neutron charge _____.

a) +1

b) -1

c) 0

d) None

Correct answer: c) 0

96. Mass of proton $\approx$ _____.

a) 1 u

b) 0 u

c) $1/1836$ u

d) 0.0005 u

Correct answer: a) 1 u

97. Mass of electron _____.

a) 1 u

b) $1/1836$ u

c) 0 u

d) Same as proton

Correct answer: b) $1/1836$ u

98. Canal rays contain _____.

a) Electrons

b) Protons

c) Neutrons

d) Alpha

Correct answer: b) Protons

99. Gold foil experiment showed _____ nucleus.

a) Large

b) Small dense positive

c) Diffuse

d) Negative

Correct answer: b) Small dense positive

100. Modern atom theory accepts _____ model.

a) Dalton

b) Thomson

c) Rutherford

d) Quantum mechanical

Correct answer: d) Quantum mechanical